

```
In [ ]: import tensorflow as tf
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.models import Model
from tensorflow.keras.preprocessing import image
import numpy as np
import matplotlib.pyplot as plt
```

```
In [3]: # Load one face images
img_path = '/home/saidul/Desktop/4-2/Deep Learning/Assignment1/data/faces/pe
img = image.load_img(img_path, target_size=(224, 224))
x = image.img_to_array(img)
x = np.expand_dims(x, axis=0)
x = tf.keras.applications.mobilenet_v2.preprocess_input(x)
```

```
In [4]: # Load pre-trained model MobileNetV2
base_model = MobileNetV2(weights='imagenet', include_top=False)

# layers for MobileNetV2 to visualize (early, middle, deep)
layer_names = [
    'Conv1',          # Early: Initial convolution
    'block_3_expand', # Middle: Expansion layer in block 3
    'block_6_expand', # Deep: Expansion layer in block 6
    'block_13_expand' # Very Deep: Expansion layer in block 13
]

outputs = [base_model.get_layer(name).output for name in layer_names]
vis_model = Model(inputs=base_model.input, outputs=outputs)

# Get activations
activations = vis_model.predict(x)
```

/tmp/ipykernel_38004/4013966172.py:2: UserWarning: `input\_shape` is undefined or non-square, or `rows` is not in [96, 128, 160, 192, 224]. Weights for input shape (224, 224) will be loaded as the default.

```
base_model = MobileNetV2(weights='imagenet', include_top=False)
1/1 ————— 1s 1s/step
```

```
In [5]: # Visualize
def plot_feature_maps(activations, layer_name, n_cols=8):
    n_features = activations.shape[-1]
    n_rows = n_features // n_cols + 1
    plt.figure(figsize=(16, n_rows*2))
    for i in range(n_features):
        plt.subplot(n_rows, n_cols, i+1)
        plt.imshow(activations[0, :, :, i], cmap='viridis')
        plt.axis('off')
    plt.suptitle(f'Feature Maps - {layer_name}', fontsize=14)
    plt.tight_layout()
    plt.show()

for i, act in enumerate(activations):
    plot_feature_maps(act, layer_names[i])
```

Feature Maps - Conv1





















